

# MLPerf(TM) Storage v3.0 - System Description

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**Division:** CLOSED  
**System name:** aws\_r5n24xlarge\_i3en12xlarge\_1c\_1s\_8h  
**Workload:** Checkpointing - llama3-8b, 8 MPI processes (CLOSED)  
**Storage system:** Alluxio Enterprise AI 3.9 (RC18) over AWS S3, via Alluxio-FUSE (POSIX)  
**Result:** Save 4.64 GiB/s / Load 5.68 GiB/s (10 checkpoints; two-invocation split)  
**Cloud:** AWS (us-west-1), single availability zone

## 1. Hardware configuration & node counts

Clients (host nodes): 1 x r5n.24xlarge

- 1 FUSE checkpoint client, 8 MPI processes
- CPU: 96 vCPU (Intel Xeon Platinum 8259CL) Memory: 768 GiB
- Storage: EBS gp3 (root) Network: 100 Gbps ENA

Storage nodes: 1 x i3en.12xlarge

- 1 Alluxio storage worker (NVMe page store) + co-located FoundationDB
- CPU: 48 vCPU (Intel Xeon Platinum 8259CL) Memory: 384 GiB
- Storage: 4x 7.5 TB NVMe SSD -> RAID0 (~28 TB) page store Network: 50 Gbps ENA
- FoundationDB: single co-located fdbserver process (configure new single memory)

## 2. System & network topology

Benchmark access is POSIX/file via Alluxio-FUSE; AWS S3 is the under-file-system (UFS). All nodes in one AWS us-west-1 availability zone over 50/100 Gbps ENA.

```
1x r5n.24xlarge  --Alluxio FUSE (POSIX, write)--> 1x i3en.12xlarge worker
(8 procs)           write-back cache              NVMe page store + FDB (metadata)
                                                         |
                                                         AWS S3 (UFS, TRANSIENT ~0)
```

## 3. Software stack & versions

- Alluxio Enterprise AI 3.9 (RC18) (distributed cache + write-back)
- Alluxio-FUSE (POSIX client)
- FoundationDB 7.3.63 (write-cache metadata)
- etcd 3.5.x (mount table + worker membership)
- Ubuntu 24.04 LTS
- OpenJDK 17
- mlpsstorage 3.0.x (uv-managed) + DLIO 3.0.x
- Backing store: AWS S3 (UFS)

## 4. Non-default configuration options

- Client write cache: alluxio.write.cache.enabled=true + FoundationDB (FdbBasedFileSystem)
- Checkpoint path routed to policyMode: TRANSIENT (write-back cache; S3 persist ~0 during run)
- Multipart write buffer: part.size=64MiB, buffer.number=512, threads=512, max.buffers.per.stream=32
- alluxio.dora.client.ufs.fallback.enabled=true; list.status.from.ufs.enabled=false
- worker.network.netty.file.transfer=TRANSFER; frame.decoder.buffer.cumulator=COMPOSITE
- FUSE JVM: -Xmx96g -XX:MaxDirectMemorySize=48g
- Two-invocation split: write (10/0) -> drop\_caches -> read (0/10); inter-phase gap <=30s (rules 4.7.1)

## 5. Notes

Simulated accelerators (no GPU). Machine-readable system parameters are in the companion systemname.yaml. CLOSED workload parameters per mlpsstorage; storage-system configuration is unconstrained in CLOSED. Dataset backing store: AWS S3.